

(a) Static Homogeneous Chunking (stragglers)

Thread 0 (P-Core):

Chunk 0 ($N/2$)

Idle

Thread 1 (E-Core):

Chunk 1 ($N/2$, slower execution)

Barrier

(b) Topology-Aware Chunking (heuristic weights)

Thread 0 (P-Core):

Chunk 0 ($\sim 0.7N$)

Thread 1 (E-Core):

Chunk 1 ($\sim 0.3N$, slower)

Barrier (Synchronized)

(c) Dynamic Dispatch (bag of tasks)

Thread 0 (P-Core):

T0

T2

T3

Wait

Thread 1 (E-Core):

T1

T4

Wait

Barrier (Synchronized)

P-Cores process tasks and segments faster; E-Cores process them slower.
Adaptive sizing or dynamic queues equalize finishing times.